

26/8/2025

CAMERAS

Slides are from Svetlana Lazebnik

Overview

- Cameras with lenses
 - Depth of field
 - Field of view
 - Lens aberrations
- Digital sensors

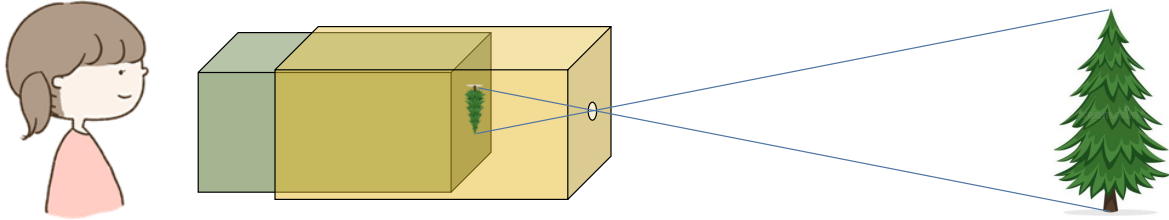
Understanding Real Cameras



Home-Made Pinhole Camera



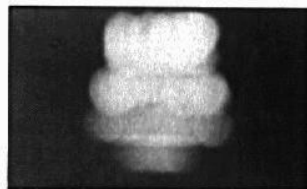
Slide by A. Efros <http://www.debevec.org/Pinhole/>



issue $\Rightarrow$ amount of light getting in is small.

Shrinking the Aperture

blur



2 mm



1 mm



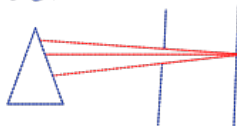
0.6 mm



0.35 mm

low
light

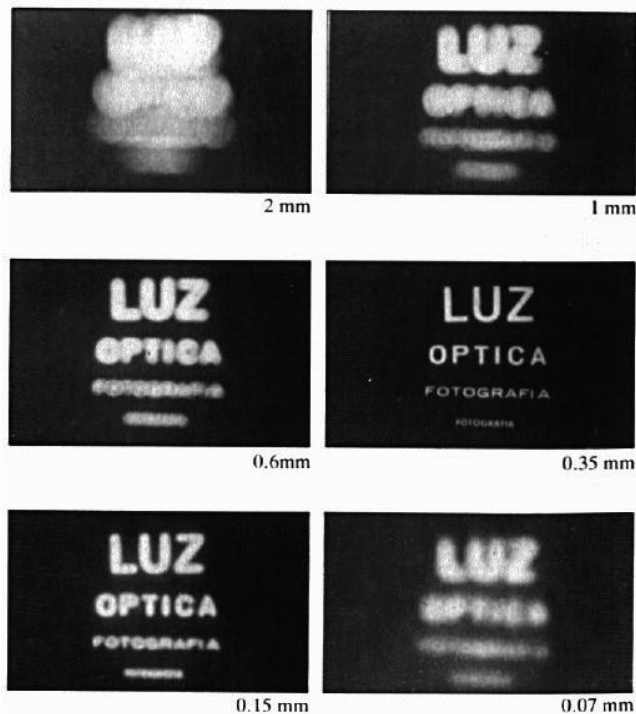
another issue
image will blur since there is no
one to one relation



Why not make the aperture as small as possible?

- Less light gets through
- Diffraction effects...

Shrinking the Aperture

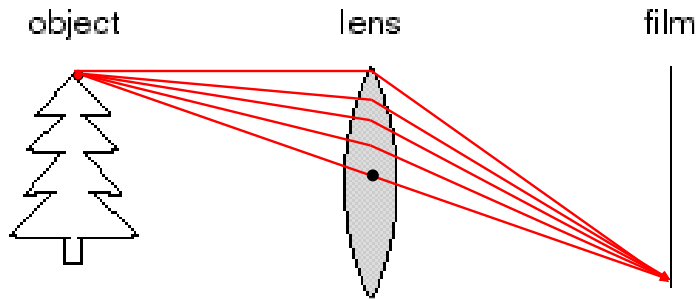


large $\Rightarrow$ blur
small $\Rightarrow$ blur and
low light

blur
again

Adding a lens

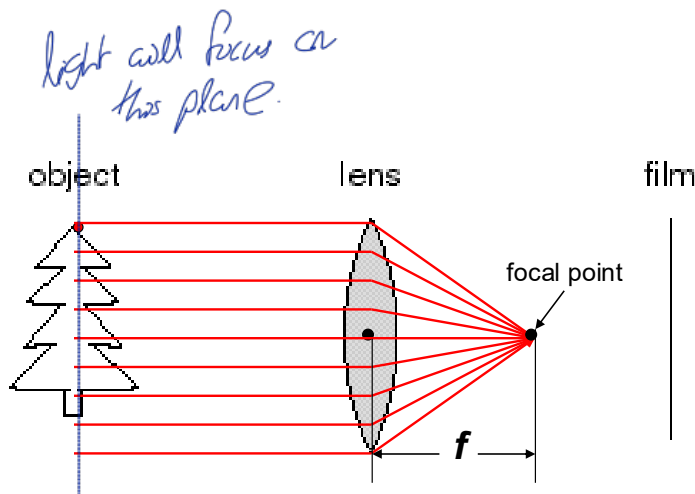
Adding a Lens



A lens focuses light onto the film

- Thin lens model:
 - Rays passing through the center are not deviated (pinhole projection model still holds)

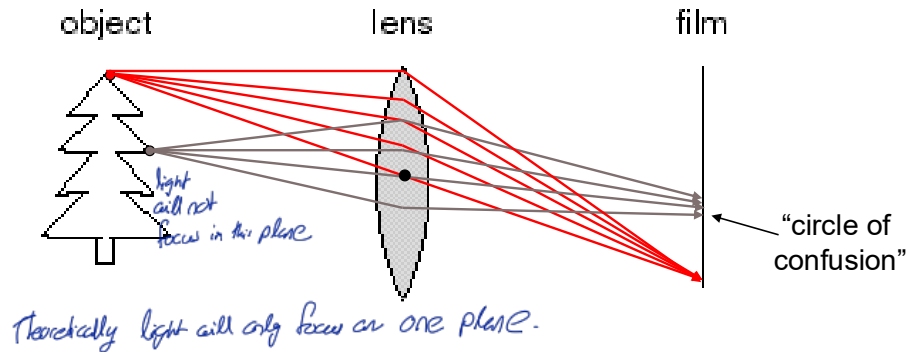
Adding a Lens



A lens focuses light onto the film

- Thin lens model:
 - Rays passing through the center are not deviated (pinhole projection model still holds)
 - All parallel rays converge to one point on a plane located at the *focal length* f

Adding a Lens

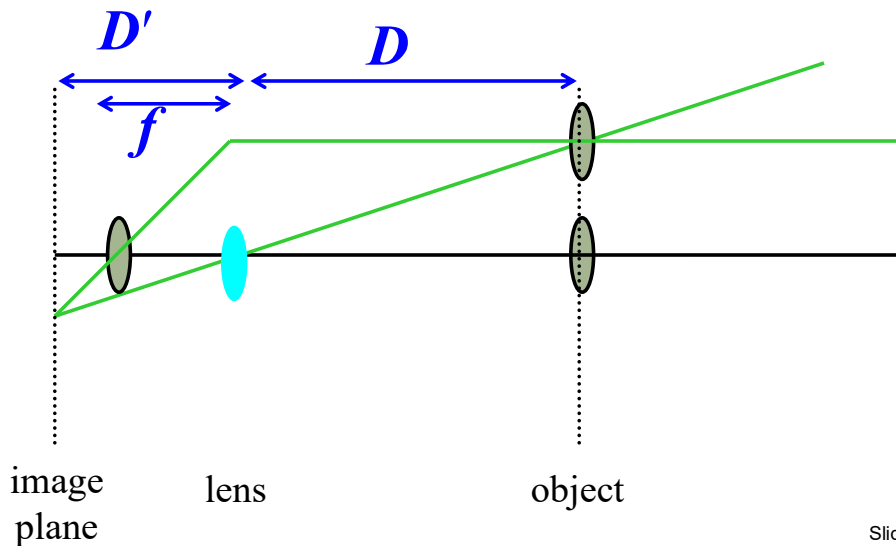


A lens focuses light onto the film

- There is a specific distance at which objects are "in focus"
 - other points project to a "circle of confusion" in the image $\Rightarrow$ issue

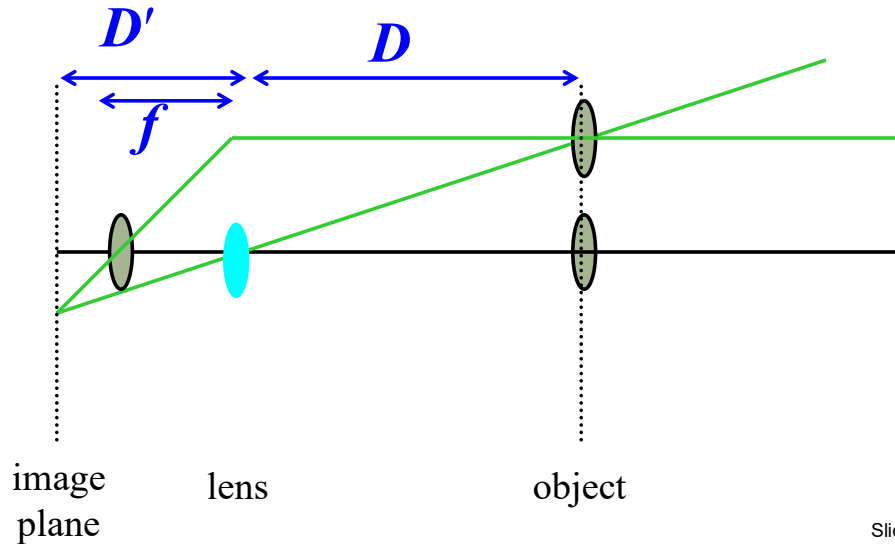
Thin Lens Formula

- What is the relation between the focal length (f), the distance of the object from the optical center (D), and the distance at which the object will be in focus (D')?



Thin Lens Formula

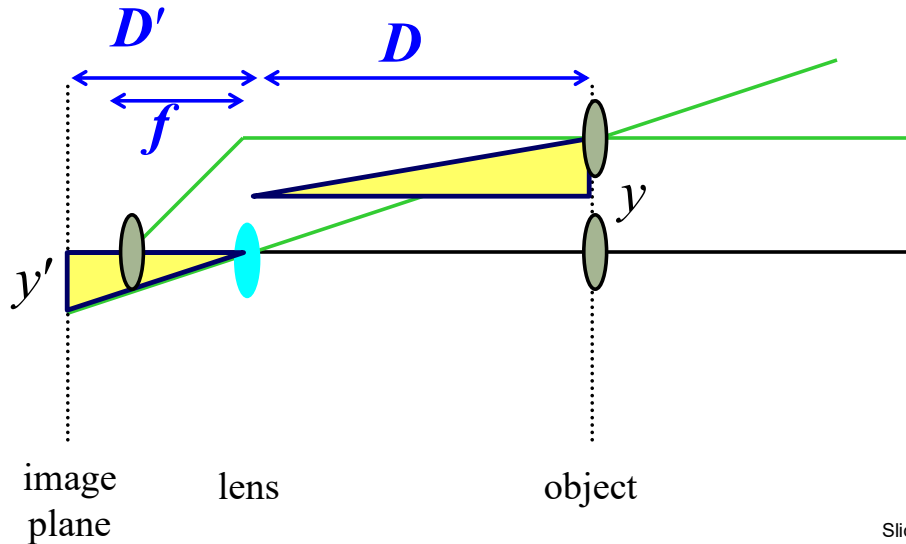
Similar triangles everywhere!



Thin Lens Formula

Similar triangles everywhere!

$$y'/y = D'/D$$

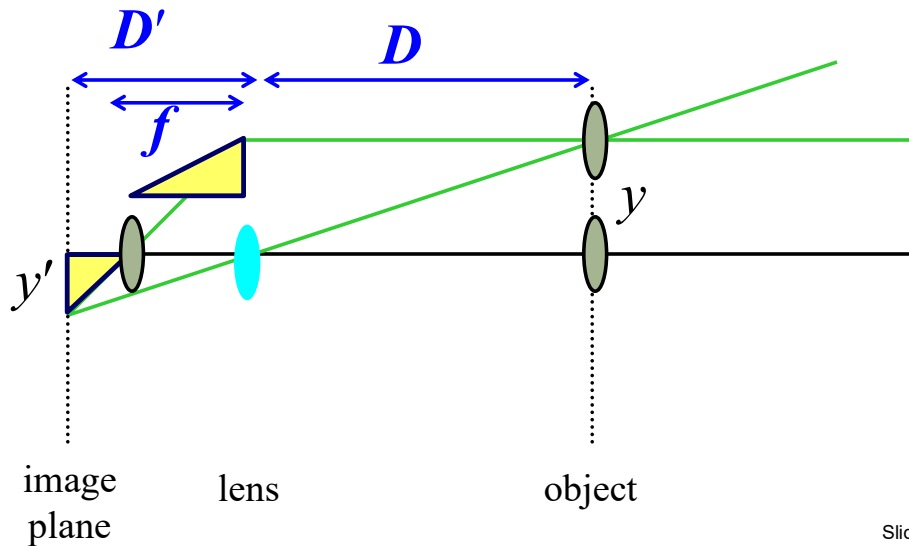


Thin Lens Formula

Similar triangles everywhere!

$$y'/y = D'/D$$

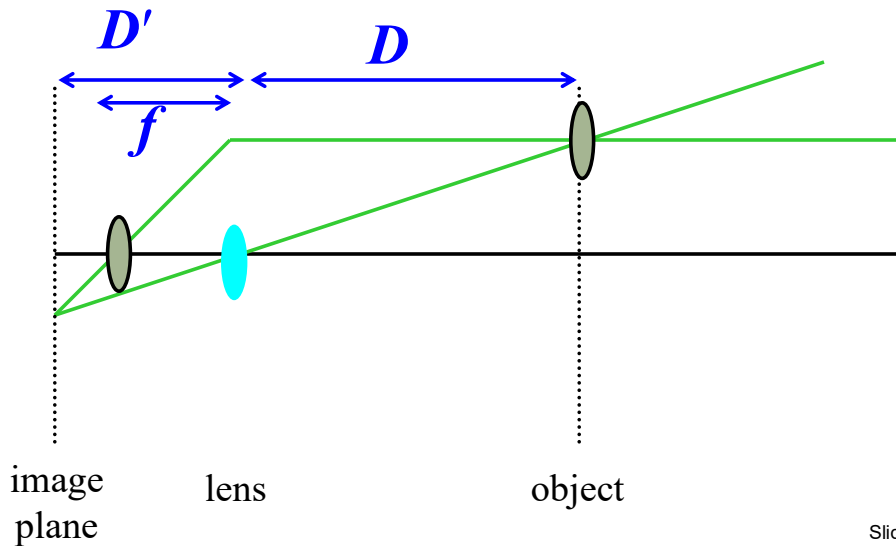
$$y'/y = (D' - f)/f$$

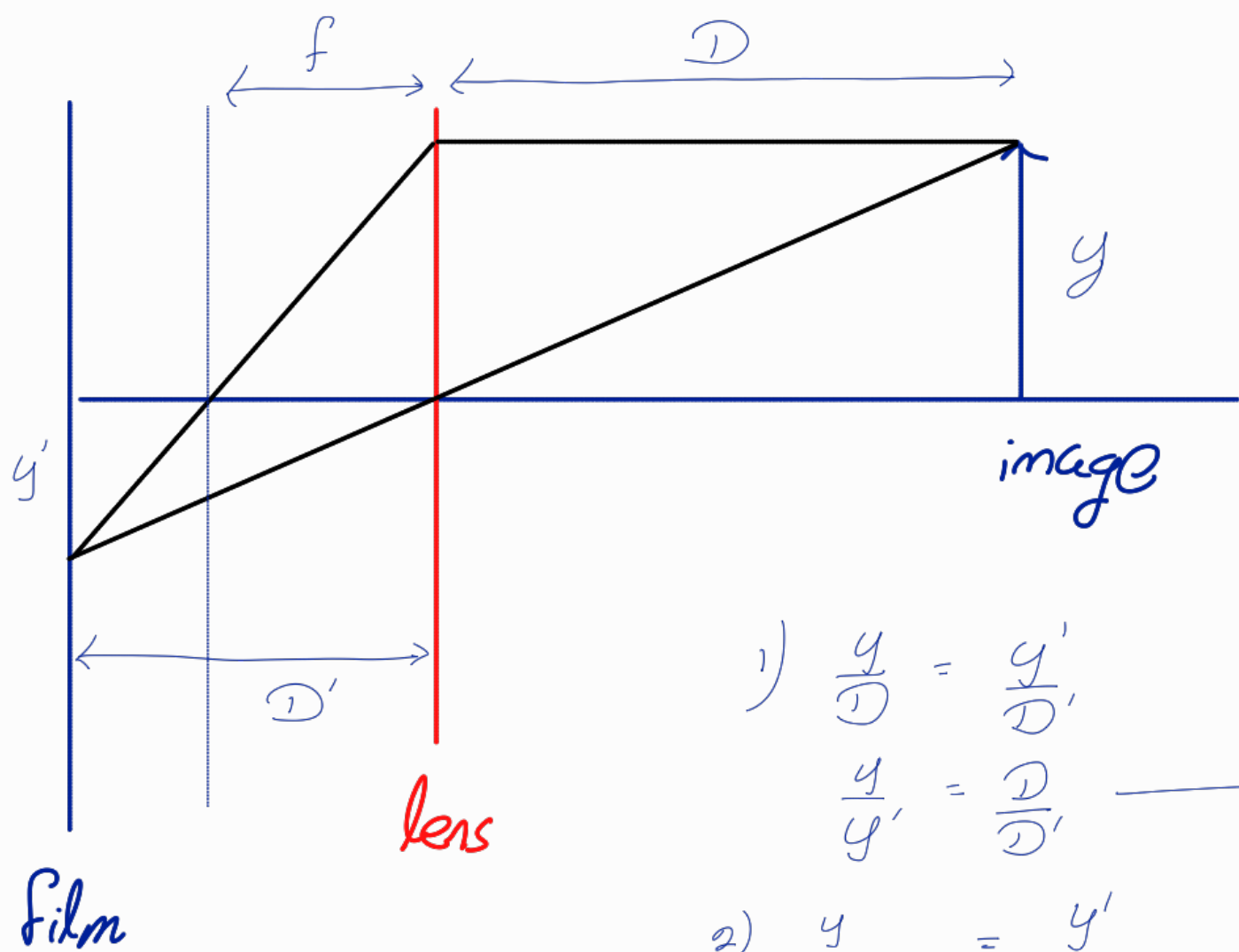


Thin Lens Formula

$$\boxed{\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}}$$

Any point satisfying the thin lens equation is in focus.





$$1) \frac{y}{D} = \frac{y'}{D'}$$

$$\frac{y}{y'} = \frac{D}{D'} \quad \text{--- (1)}$$

$$2) \frac{y}{f} = \frac{y'}{D' - f}$$

$$\frac{y}{y'} = \frac{f}{D' - f} \quad \text{--- (2)}$$

$$(1) = (2)$$

$$\frac{D}{D'} = \frac{f}{D' - f}$$

$$DD' - fD = fD'$$

$$\frac{1}{f} - \frac{1}{D'} = \frac{1}{D}$$

$$\frac{1}{D} + \frac{1}{D'} = \frac{1}{f}$$

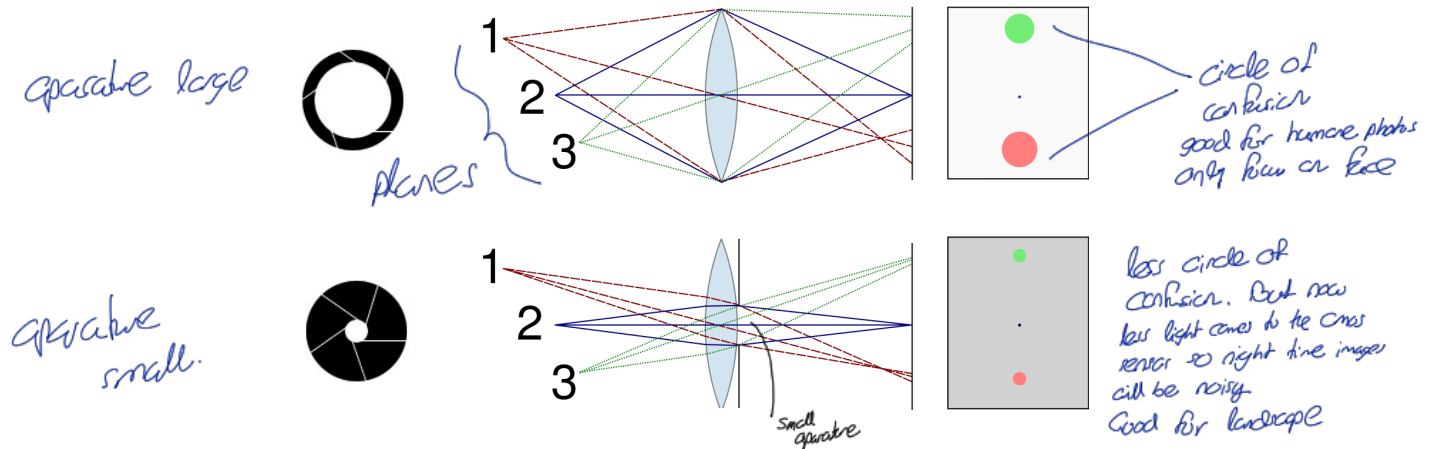
Depth of Field



<http://www.cambridgeincolour.com/tutorials/depth-of-field.htm>

Slide by A. Efros

Controlling Depth of Field



Changing the aperture size affects depth of field

- A smaller aperture increases the range in which the object is approximately in focus
- But small aperture reduces amount of light – need to increase exposure

http://en.wikipedia.org/wiki/File:Depth_of_field_illustration.svg

Varying the Aperture



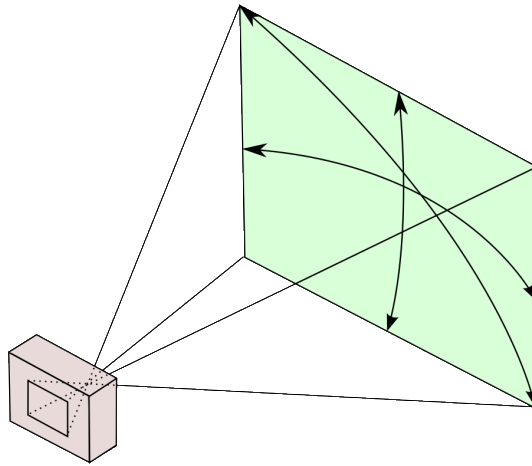
Large aperture = small DOF



Small aperture = large DOF

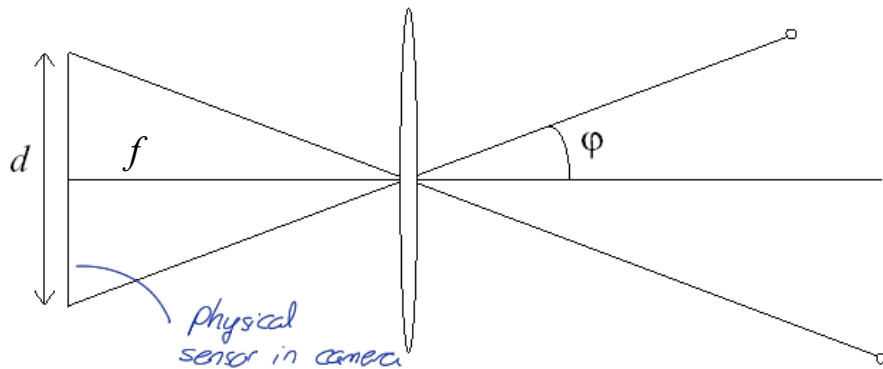
Field of view

- The field of view is the angular extent of the world observed by the camera ([Wikipedia](#))
- What determines the FOV?



Field of view

- The field of view is the angular extent of the world observed by the camera ([Wikipedia](#))
- What determines the FOV?
 - Focal length (f), length of the sensor (d):

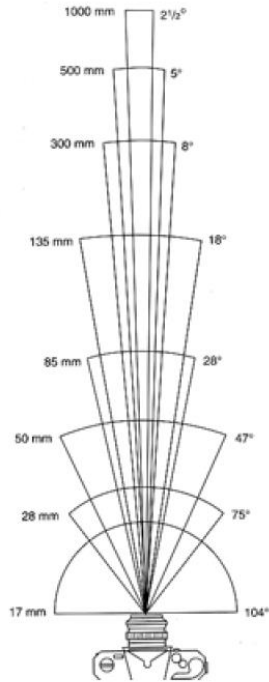


$$\varphi = \tan^{-1} \frac{d}{2f}$$

- Larger focal length = smaller FOV

The light needs to focus on sensor

Field of View



17mm



28mm



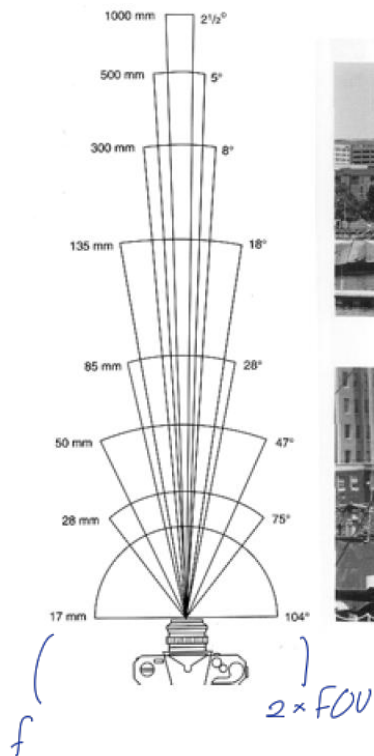
50mm



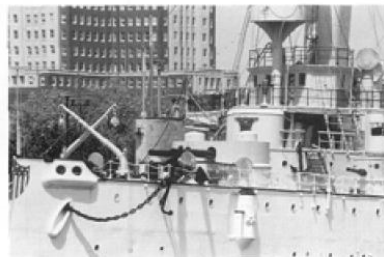
85mm

Slide by A. Efros

Field of View



135mm



300mm

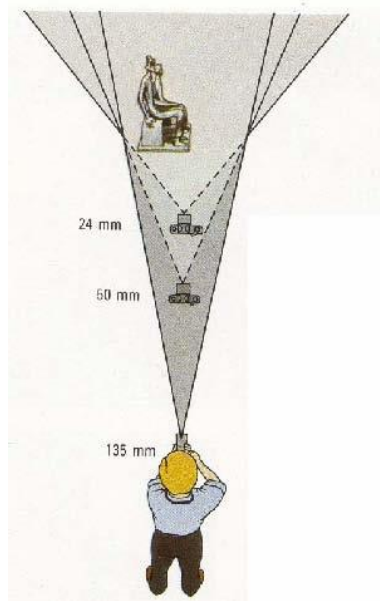


135mm



300mm

Field of View / Focal Length



For large FOV cameras we need to be close to the object to fit to frame. Then the close distance clearly shows size diff between near and far features giving a 3D appearance



Large FOV, small f
Camera close to car

24mm
3D-native can be seen



Small FOV, large f
Camera far from the car

135mm
image is flat

Sources: A. Efros, F. Durand

Same Effect for Faces

for faces (85-105) shot telephoto is best

(14-35)mm

(50mm)

(135+)mm



wide-angle

small f
large FOV



standard

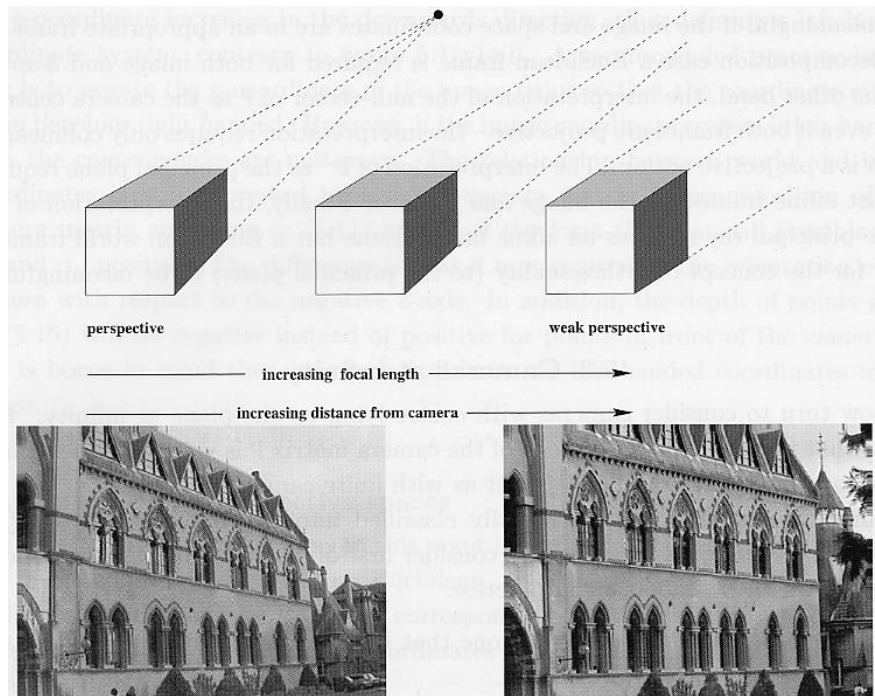


telephoto

large f
small FOV

Source: F. Durand

Approximating an Orthographic Camera



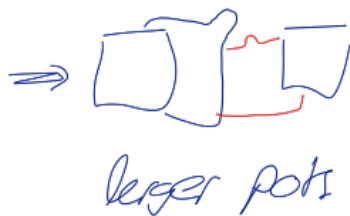
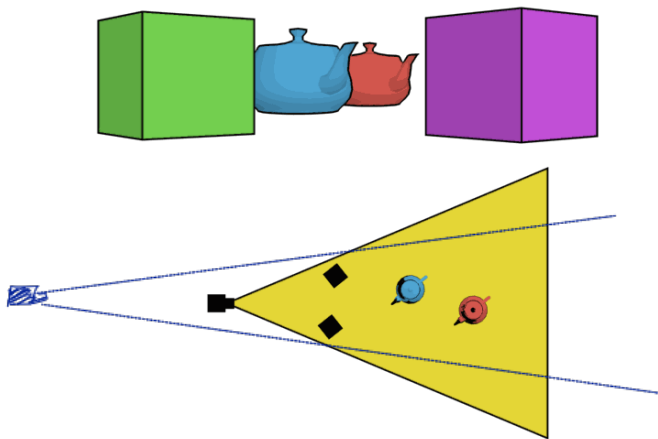
large f
↓
small FOV
↓
camera must be far
↓
flat image

long focal
length.
parallel lines are
parallel (flat image)

Source: Hartley & Zisserman

The Dolly Zoom

- Continuously adjusting the focal length while the camera moves away from (or towards) the subject



http://en.wikipedia.org/wiki/Dolly_zoom

Given example

$\Rightarrow$ moving the camera forward while
reducing focal length
(zoom out)

Result is $\Rightarrow$ foreground remains same
while more of background
comes to view

The Dolly Zoom

- Continuously adjusting the focal length while the camera moves away from (or towards) the subject
- “The Vertigo shot”



[Example of dolly zoom from Goodfellas](#) (YouTube)

[Example of dolly zoom from La Haine](#) (YouTube)

Overview

- Cameras with lenses
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 - Lens aberrations

look into light painting

Exposure triangle: shutter speed / ISO / aperture

trade off

how to measure

regular values

scenarios

aperture $\propto \frac{1}{\text{Diameter}}$?

Shutter speed $\Rightarrow$ how long camera sensor is exposed to light

slow (longer) $\Rightarrow$ bright image

fast $\Rightarrow$ freeze shots

but more light needed

ISO $\Rightarrow$ sensor sensitivity to light

Low $\Rightarrow$ need more light but clean photo

high $\Rightarrow$ shooting in dark but noisy

Aperture $\Rightarrow$ opening in lens

narrow $\Rightarrow$ keep stuff sharp (or blur)

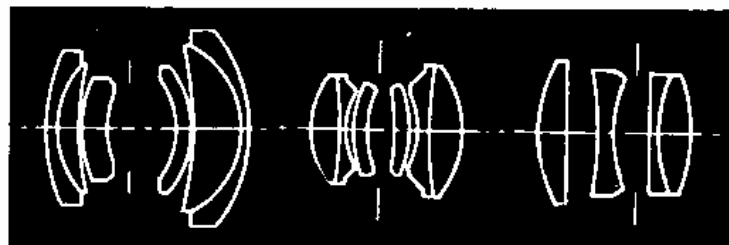
large $\Rightarrow$ blur background

Aperture is measured by f-number

$$\text{f-number} = \frac{f}{D}$$

example 50mm lens at $\frac{f}{2} \Rightarrow D = \frac{50}{2} = 25$
f-number = 2

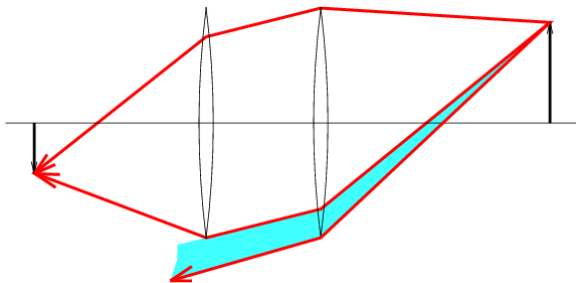
Real Lenses



Usually we keep iso as low as possible and change the other parameters.

Lens Flaws: Vignetting

1 Darkening of image corners compared to center

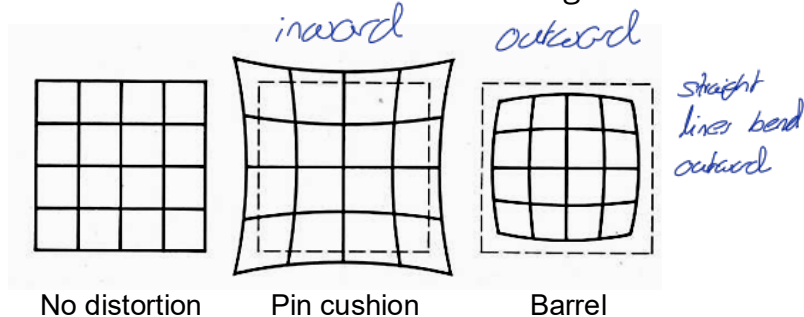


light hitting the edge of sensor has to travel far and also passes in the center of sensor less straight at incoming light making it more difficult at collecting light



Lens Flaws: Radial Distortion

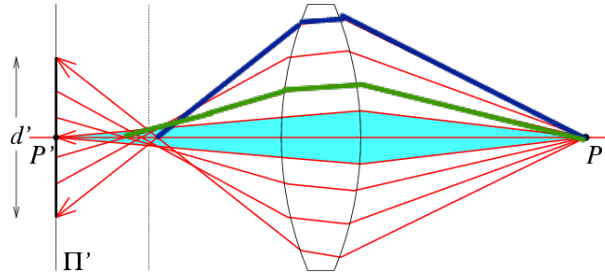
- Caused by **imperfect lenses**
- Deviations are most noticeable near the edge of the lens



Lens flaws: Spherical Aberration

Spherical lenses don't focus light perfectly

Rays farther from the optical axis focus closer

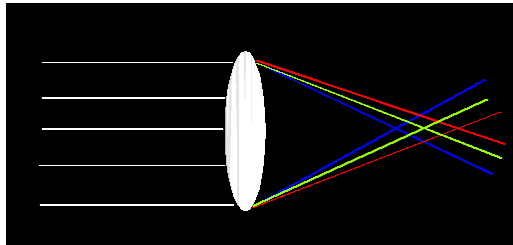


Lens Flaws: Chromatic Aberration

Lens has different refractive indices for different wavelengths: causes color fringing

unwanted colored edges that appear along boundaries

*glass refract (bends)
different wavelengths
by diff amounts
⇒ dispersion*



*RGB bend
differently.*

Near Lens Center



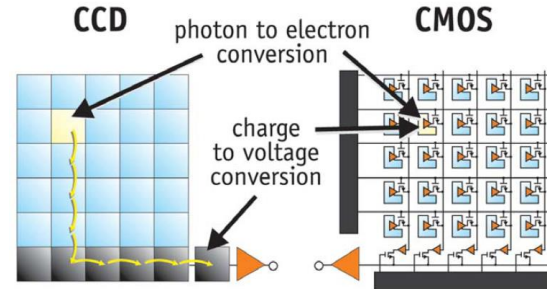
Near Lens Outer Edge



Overview

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- Digital sensors

Digital Camera



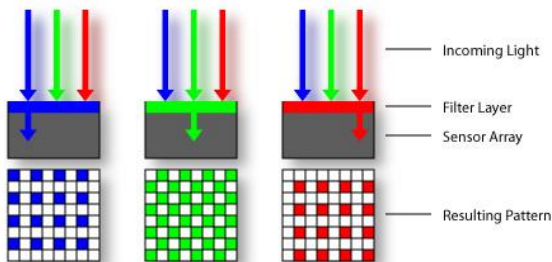
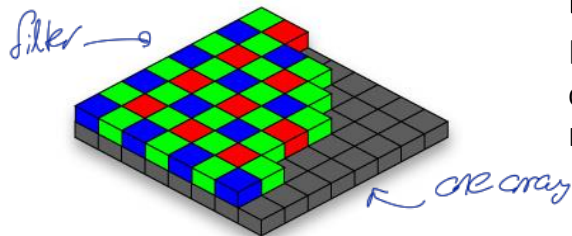
CCDs move photogenerated charge from pixel to pixel and convert it to voltage at an output node. CMOS imagers convert charge to voltage inside each pixel.

A digital camera replaces film with a sensor array

- Each cell in the array is light-sensitive diode that converts photons to electrons
- Two common types
 - **Charge Coupled Device (CCD)**
 - **Complementary metal oxide semiconductor (CMOS)**
- <http://electronics.howstuffworks.com/digital-camera.htm>

Color filter arrays

Bayer grid (1976)



$R \Rightarrow 25\%$
 $G \Rightarrow 50\%$
 $B \Rightarrow 25\%$

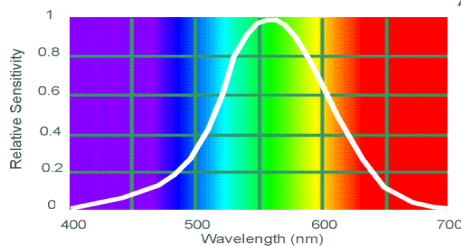
Demosaicing:

Estimation of missing components from neighboring values



interpolate

Why more green?



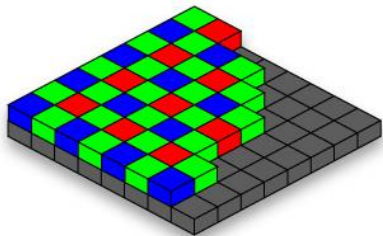
Human Luminance Sensitivity Function

Source: Steve Seitz

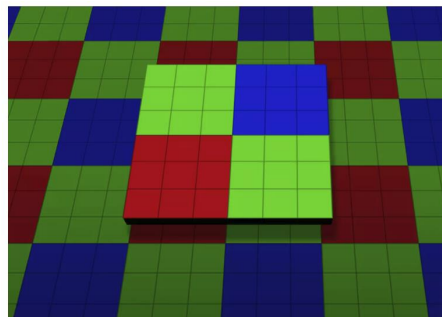
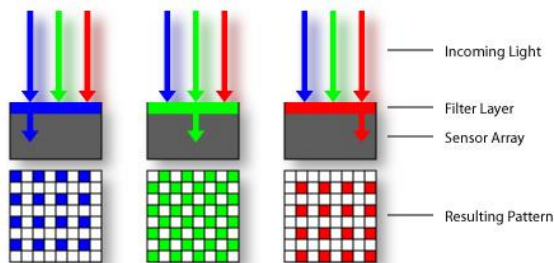
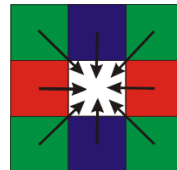
human eye is more sensitive to green more green to capture more luminance detail which is what our visual system prioritize

Color filter arrays

Bayer grid (1976)



Demosaicing:
Estimation of missing components from neighboring values



Recent cameraphone technology: pixel binning

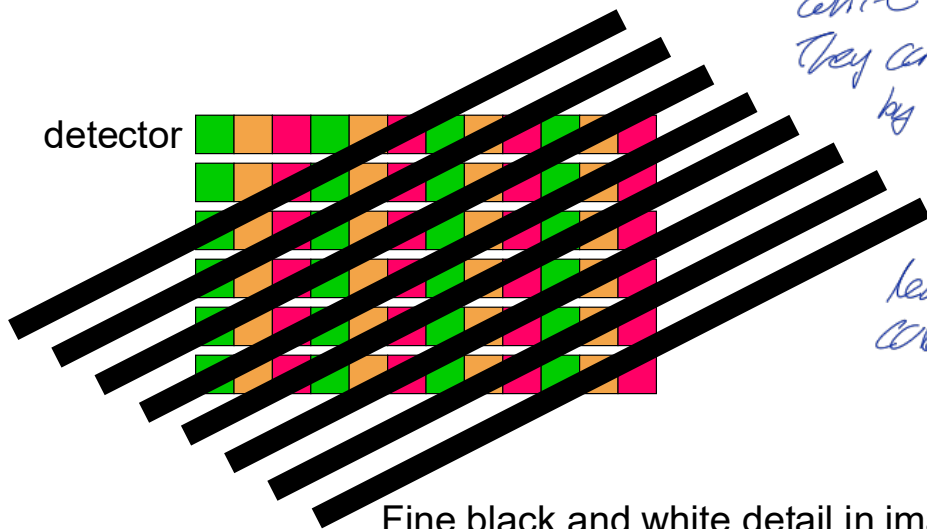
Problem with Demosaicing: Color Moire



Slide by F. Durand

The Cause of Color Moire

fake colours that appear when there are fine black and white details. They cannot be captured by the grid sampling rate so it is undersampled leading to false colours



Fine black and white detail in image misinterpreted as color information

Misc. digital camera artifacts

Noise

- low light is where you most notice noise
- light sensitivity (ISO) / noise tradeoff
- stuck pixels



In-camera processing

- oversharpening can produce halos



Compression

- JPEG artifacts, blocking



Blooming

- CCD charge overflowing into neighboring pixels

Color artifacts

- Color moire
- Purple fringing from microlenses



Digital Camera Artifacts

Noise

- low light is where you most notice [noise](#)
- light sensitivity (ISO) / noise tradeoff
- stuck pixels



In-camera processing

- oversharpening can produce [halos](#)



Compression

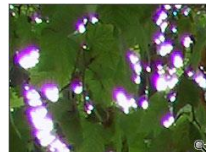
- JPEG artifacts, blocking

Blooming

- charge [overflowing](#) into neighboring pixels

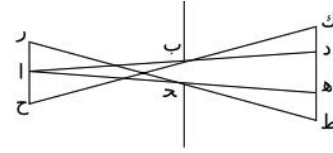
Color artifacts

- [purple fringing](#) from microlenses,
- white balance

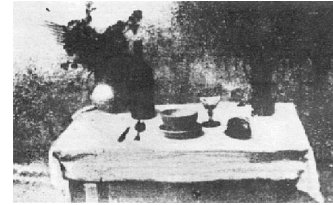


Historic Milestones

- **Pinhole model:** Mozi (470-390 BCE), Aristotle (384-322 BCE)
- **Principles of optics (including lenses):** Alhacen (965-1039 CE)
- **Camera obscura:** Leonardo da Vinci (1452-1519), Johann Zahn (1631-1707)
- **First photo:** Joseph Nicéphore Niépce (1822)
- **Daguerréotypes** (1839)
- **Photographic film** (Eastman, 1889)
- **Cinema** (Lumière Brothers, 1895)
- **Color Photography** (Lumière Brothers, 1908)
- **Television** (Baird, Farnsworth, Zworykin, 1920s)
- **First consumer camera with CCD**
Sony Mavica (1981)
- **First fully digital camera:** Kodak DCS100 (1990)



Alhacen's notes



Niepce, "La Table Servie," 1822



Old television camera

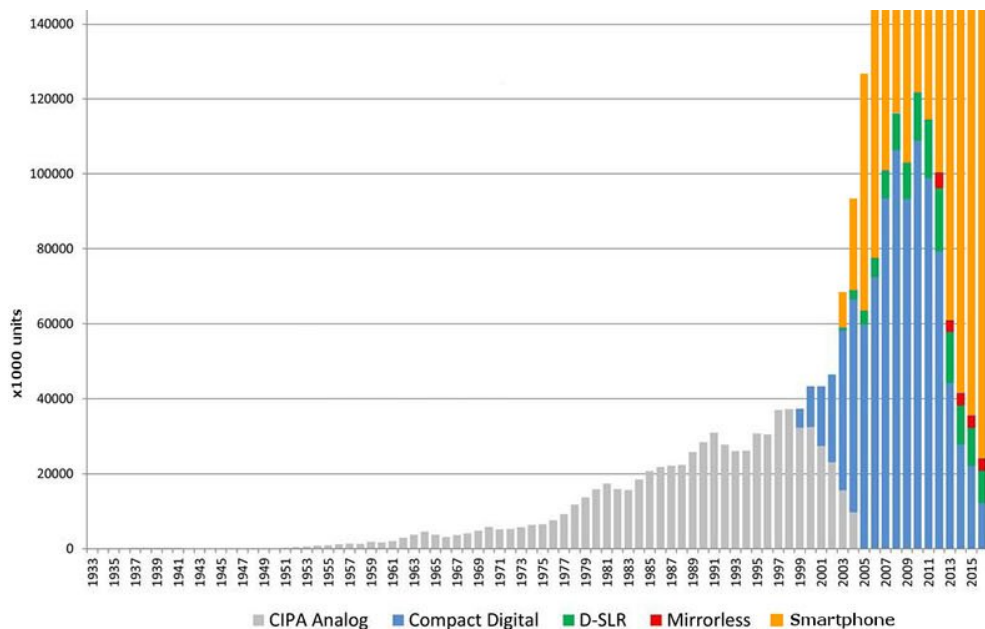
First Digitally Scanned Photograph

- 1957, 176x176 pixels



<http://listverse.com/history/top-10-incredible-early-firsts-in-photography/>

Camera Sales Over Time



[Source](#)

Camera Sales Over Time

The full chart...



More information on color and cameras:
https://www.eecs.yorku.ca/~mbrown/ICCV19_Tutorial_MSBBrown.pdf

[Source](#)